ORIGINAL ARTICLE

Efficacy and Safety of extended-release oral opioid for cough in idiopathic pulmonary fibrosis: A systematic review and meta-analysis of randomized controlled trials

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ABSTRACT

Background and aim: Patients with idiopathic pulmonary fibrosis (IPF) experience a high burden of cough. Currently, no established effective therapy exists for cough in IPF. The overall evidence base of extended-release (ER) oral opioid therapies in this population remains limited. This study aimed to compare the efficacy and safety of ER oral opioids versus placebo for cough in patients with IPF.
Methods: PubMed, Embase, Cochrane, and ClinicalTrial.gov databases were reviewed for randomized controlled trials (RCTs). The primary outcome was 24-hour objective cough frequency. Secondary outcomes included Evaluating Respiratory Symptoms-Idiopathic Pulmonary Fibrosis (E-RS IPF) cough subscale score, Cough Severity-Numeric Rating Scale (CS-NRS) score. Safety outcomes included serious adverse events and adverse events leading to discontinuation.
Results: Three RCTs, including 250 patients, were analyzed. Opioid interventions included ER nalbuphine and morphine. ER oral opioid statistically significantly improved 24-hour objective cough frequency (GMR 0.46; 95% CI 0.23 to 0.92; p = 0.029), CS-NRS score (MD -1.69; 95% CI -2.55 to -0.84; p < 0.001), and E-RS IPF cough

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subscale score (MD -0.49; 95% CI -0.89 to -0.09; p = 0.016), compared with placebo. No statistically significant differences were observed for serious adverse events or adverse events leading to discontinuation.
Conclusions: ER oral opioids significantly improved cough frequency and patient-reported cough severity and discomfort in patients with IPF, without a significant increase in serious adverse events or treatment discontinuation. Extended-release oral opioids may be cautiously considered as a therapeutic option for cough in patients with IPF.

Key words: cough, idiopathic pulmonary fibrosis, opioids

Introduction

Idiopathic pulmonary fibrosis (IPF) is a chronic and progressive interstitial lung disease (ILD) (1). A substantial proportion of patients experience chronic and refractory cough that markedly impairs health-related quality of life (QoL) (2,3), and severe cough has been linked to faster disease progression and poorer survival (4). The pathophysiological mechanisms underlying cough in IPF remain incompletely understood, which has limited the development of effective therapeutic options (5). Proposed mechanisms underlying cough in IPF include increased cough reflex sensitivity due to mechanical stimulation from fibrosis and heightened sensory neuronal responsiveness (6,7). The evidence remains limited that antifibrotic therapies meaningfully improve cough symptoms (8–10). Previous investigations into alternative therapies for IPF-associated cough, such as inhaled cromoglycate, proton pump inhibitors, and prednisone, have also failed to demonstrate a significant reduction in cough frequency (11–13). Consequently, there remains no established effective therapy for cough in IPF (14).

In this context, ER oral opioid therapies have recently been studied as a potential treatment for chronic cough in IPF (5,15,16). Opioids mainly suppress cough through μ- and κ-opioid receptors distributed along central and peripheral components of the cough reflex pathway, likely reducing heightened sensory neuronal responsiveness causing cough in IPF (17). In a randomized crossover trial, low-dose controlled-release morphine reduced objectively measured awake cough frequency and improved QoL (5). Other trials of extended-release (ER) oral nalbuphine demonstrated a reduction in objectively measured cough frequency (15,16). However, side effects such as nausea, constipation, and dizziness were reported more frequently in the opioid group in these trials (5,15,16). In addition, the evaluation of serious adverse events was limited by the small sample size and relatively short follow-up duration (5,15).

To date, the number of available RCTs remains limited, and the overall efficacy and safety of ER oral opioid therapy for cough in IPF remain uncertain. Therefore, we conducted a systematic review and meta-analysis to systematically evaluate the efficacy and safety of ER oral opioid therapies for chronic cough in patients with IPF.

Methods

This systematic review and meta-analysis were performed following the Cochrane Collaboration Handbook for Systematic Review of Interventions and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (18). Additionally, our study protocol was registered using the following registration number CRD420261329246 in the International Prospective Register of Systematic Reviews.

Eligibility criteria

We included studies that met all the following eligibility criteria: (1) RCTs; (2) comparison of ER oral opioid therapies with placebo; (3) enrollment of patients with IPF; and (4) reporting of at least one of the clinical outcomes of interest. Citations were excluded if: (1) patient populations overlapped with those of other included studies and (2) were abstract from conferences. There were no restrictions on the follow-up duration or language. Detailed eligibility criteria for each included study are provided in Supplementary Table 1.

Search strategy and data extraction

We systematically searched PubMed, Embase, Cochrane Central Register of Controlled Trials, and ClinicalTrials.gov from inception to February 28, 2026. The complete search strategy is presented in Supplementary Table 2.

References from all included studies, previous systematic reviews, and meta-analyses were manually searched for additional studies. Two authors independently screened and extracted data using dedicated spreadsheets. Any discrepancies were resolved through consultation with a senior author. We extracted data on study characteristics, participant characteristics, intervention details, and outcomes of interest. When data were missing or unclear, we contacted the study authors where possible or estimated values using established methods.

Endpoints

The primary endpoint was 24-hour objective cough frequency. Secondary outcomes included Evaluating Respiratory Symptoms-Idiopathic Pulmonary Fibrosis (E-RS IPF) cough subscale score, Cough Severity-Numeric Rating Scale (CS-NRS) score. Safety outcomes included serious adverse events, adverse events leading to discontinuation, nausea, vomiting, constipation, somnolence, fatigue, dizziness, headache, and dry mouth. The definitions of each endpoint used in the included studies are provided in Supplementary Table 3. The Leicester Cough Questionnaire (LCQ) was included as an additional post-hoc secondary efficacy endpoint.

Assessment of the risk of bias and quality of evidence

Two authors independently assessed the risk of bias and disagreements were resolved with the senior author. Studies were appraised using the Cochrane Collaboration's tool for assessing risk of bias in randomized trials (RoB-2), which evaluates five domains: (1) randomization process, (2) deviations from intended protocol, (3) missing outcome data, (4) measurement of the outcome, and (5) selection of the reported result (19). Additionally, we employed the RoB-2 tool for crossover trials for relevant studies, which incorporates specific assessments for bias arising from period and carryover effects. The quality of evidence was assessed using the Grading of Recommendations Assessment Development and Evaluation (GRADE) approach by two independent reviewers and disagreements were resolved with the third author (20). We did not assess publication bias due to the limited number of studies included ($n < 10$), as regression-based methods lack sufficient power to distinguish chance from true asymmetry (21).

Statistical analysis and sensitivity analyses

Binary outcomes were reported as odds ratios (ORs) using the Mantel–Haenszel method, and continuous outcomes were reported as geometric mean ratio (GMR) or mean differences (MD), each with their respective 95% confidence intervals (95% CI). A random-effects model was used to account for potential between-study variability. Restricted maximum likelihood (REML) was applied to estimate the between-study variance (τ^2). Heterogeneity was assessed using the I^2 statistic and the Cochrane Q test; an I^2 value ≥25% and a Cochrane Q test p-value <0.10 were considered suggestive of substantial heterogeneity. All analyses were conducted using RStudio (version 4.4.2).

To enable pooling with geometric mean ratios (GMRs), studies reporting arithmetic means and standard deviations were first transformed into log-transformed means and standard errors, following the approach described by Higgins et al. (2009) (22). After standardization, these estimates were converted into GMRs and pooled in the meta-analysis.

For cross-over trials, to avoid unit-of-analysis errors in continuous outcomes, we followed the recommendations of the Cochrane Handbook (23). The correlation coefficient was estimated using a parallel-group trial as a reference. This coefficient was then used to calculate the standard deviation of

Table 1. Baseline patient and study characteristics.

Study and year	Type of study	Intervention	Regime	Sample size (I/C)	Follow-up	Male	Age, years	24-hour cough frequency†	Daytime cough frequency†	LCQ	AFT	PPI	Oxygen therapy
PACIFY COUGH, 2024(5)	Crossover RCT	Controlled-release Morphine	5 mg twice daily	44/44	5 weeks	31 (70.0)	71	NA	21.6	13.2 (0.5)	26 (59.0)	13 (30.0)	4 (9.0)
Maher, 2023(15)	Crossover RCT	Nalbuphine ER	27 mg once daily, titrated up to 162 mg twice daily	38/38	7 weeks	32 (84.2)	74	21.2	28	NA	18 (47.4)	28 (68.3)	NA
CORAL, 2026*(16)	Parallel RCT	Nalbuphine ER	27, 54, or 108 mg twice daily	125/40	8 weeks	29 (69.0)	69	24.6	NA	12.4 (4.0)	31 (73.8)	21 (50.0)	6 (14.3)

All population was IPF and control was placebo. *Abbreviations*: AFT: antifibrotic therapy; C: control; DLCO: diffusing capacity of the lung for carbon monoxide; FVC: forced vital capacity; I: intervention; ILD: interstitial lung disease; IPF: idiopathic pulmonary fibrosis; LCQ: Leicester Cough Questionnaire; PPI: proton pump inhibitor; RCT: randomized controlled trial. Categorical data were reported as counts and frequencies (%). Continuous data reported as mean or median. AFT, PPI, and Oxygen therapy were reported as the number of patients received these treatments. *Baseline characteristics for the CORAL study were extracted from the 27-mg twice-daily dose group. †Cough frequency is reported as coughs per hour the within-participant difference and the corresponding standard error for the cross-over trials. For studies with multiple intervention arms, treatment groups were combined into a single group to avoid double-counting of the control group (23). These calculations were performed using the Review Manager (RevMan) online calculator. A full disclosure of the statistical analyses can be found on the Supplementary Methods.

Results

The initial search yielded 224 results on February 28, 2026. After removing duplicate results and applying the eligibility criteria, 39 records were selected for the full-text review, as detailed in Figure 1. Of these, three RCTs were included in this systematic review and meta-analysis (4,14,15). A total of 250 participants were included. Mean age ranged from 69 to 74 years, with 69 to 84% male. ER oral opioid regimens included low dose controlled-release morphine 5 mg administered twice daily; ER nalbuphine initiated at 27 mg once daily with titration up to 162 mg twice daily; and fixed-dose regimens of 27, 54, or 108 mg twice daily. Median follow-up was 26 days. Baseline 24-hour objective cough frequency ranged from 21.2 to 24.6 coughs per hour. The baseline characteristics of the participants are presented in Table 1

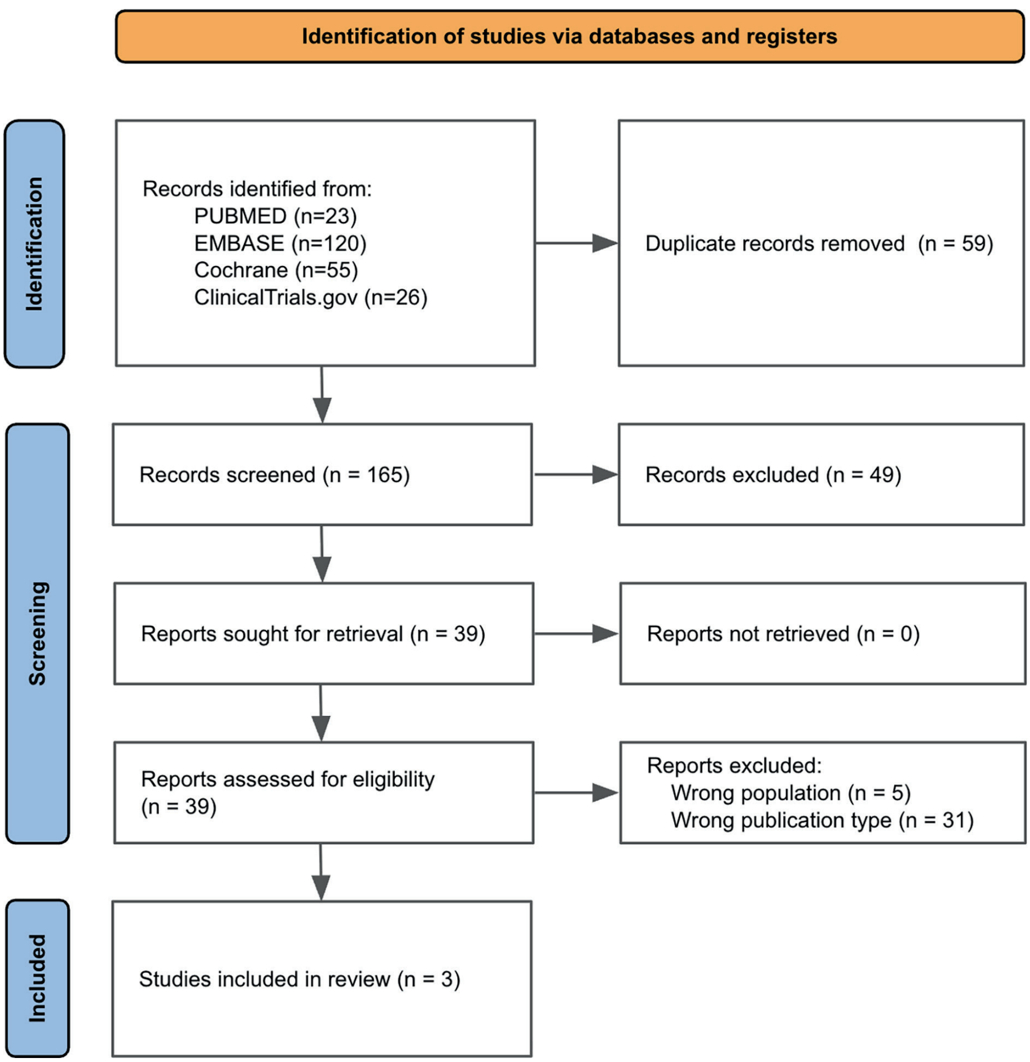

Figure 1. PRISMA Flow Chart.

and Supplementary Table 4. Quantitative synthesis differed by outcome domain. The prespecified pooled efficacy analyses were based on trials evaluating ER oral nalbuphine because efficacy outcomes suitable for direct quantitative synthesis were available only from the nalbuphine trials. An additional post-hoc analysis of LCQ incorporated trials evaluating both ER nalbuphine and low-dose controlled-release morphine. Safety outcomes were synthesized using both oral opioids because extractable safety data were available from all eligible trials.

Pooled analysis of all studies and sensitivity analyses

ER oral opioid statistically significantly reduced objective 24-hour cough frequency (GMR 0.46; 95% CI 0.23 to 0.92; $p = 0.029$; $I^2 = 45.6\%$; Figure 2) and improved CS-NRS score (MD -1.69; 95% CI -2.55 to -0.84; $p < 0.001$; $I^2 = 62.1\%$; Figure 3), E-RS IPF cough subscale score (MD -0.49; 95% CI -0.89 to -0.09; $p = 0.017$; $I^2 = 72.6\%$; Figure 4), and LCQ score (MD 1.80; 95% CI 0.58 to 3.01; $p = 0.004$; $I^2 = 49.1\%$; Supplementary Figure 7) compared with placebo.

The GMR of 0.46 corresponds to a 54% reduction in objective 24-hour cough frequency, exceeding the proposed ≥30% meaningful change threshold for 24-hour cough frequency in chronic cough (24). For the CS-NRS, no IPF-specific minimal clinically important difference (MCID) has been established; using the chronic cough visual analog scale threshold referenced in the CORAL trial, approximately equivalent to a 3-point change on a 0-10 CS-NRS, the pooled MD did not reach this reference threshold (16). For the E-RS IPF cough subscale, no established MCID has been clearly defined. For the

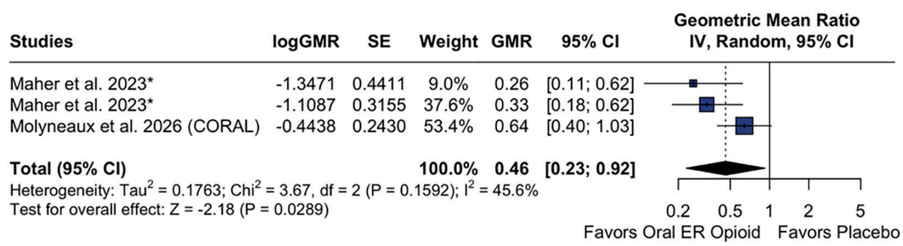

Figure 2. Forest plot for 24-hour objective cough frequency.*Maher et al. appears twice because the trial used a randomized cross-over design with two treatment periods; period-specific treatment comparisons were included separately in the meta-analysis.

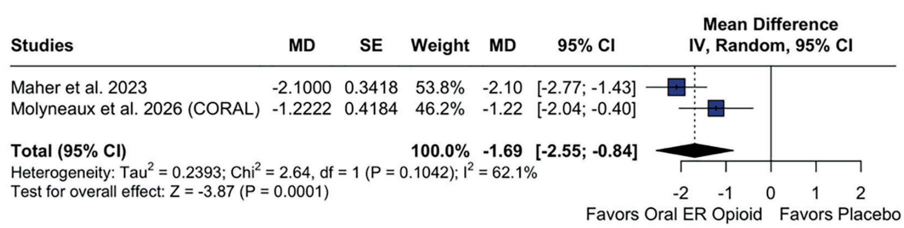

Figure 3. Forest plot for CS-NRS score.

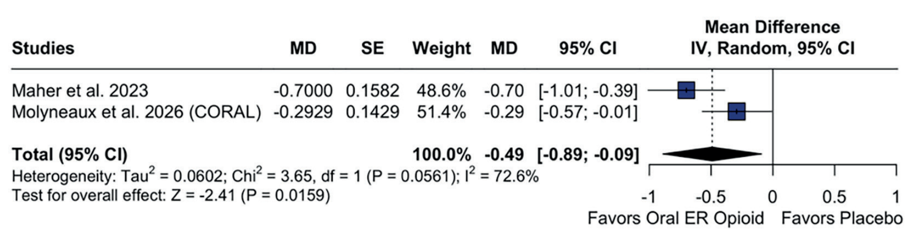

Figure 4. Forest plot for E-RS IPF cough subscale score.

LCQ, an MCID of 1.3 points has been established in chronic cough, and the pooled MD exceeded this threshold (25).

All safety endpoints are displayed at Figure 5. The risk of serious adverse events did not clearly differ between ER oral opioids and placebo, although the

Studies	Oral ER Opioids Events	Oral ER Opioids Total	Placebo Events	Placebo Total	Weight	OR	95% CI
Serious AE							
Molyneaux et al. 2026 (CORAL)	2	125	4	40	59.8%	0.15	[0.03; 0.83]
Wu et al. 2024, (PACIFY COUGH)	0	43	1	43	17.3%	0.33	[0.01; 8.22]
Maher et al. 2023	1	42	1	42	22.9%	1.00	[0.06; 16.53]
Total (95% CI)	**3**	**210**	**6**	**125**	**100.0%**	**0.26**	**[0.07; 1.00]**
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 1.32$, df = 2 (P = 0.5156); $I^2 = 0\%$							
Test for overall effect: Z = -1.96 (P = 0.0501)							
AE discontinuation							
Molyneaux et al. 2026 (CORAL)	7	125	2	40	49.1%	1.13	[0.22; 5.66]
Wu et al. 2024, (PACIFY COUGH)	1	43	0	43	23.5%	3.07	[0.12; 77.50]
Maher et al. 2023	9	42	0	42	27.4%	24.10	[1.35; 429.25]
Total (95% CI)	**17**	**210**	**2**	**125**	**100.0%**	**3.30**	**[0.51; 21.51]**
Heterogeneity: $Tau^2 = 1.1849$; $Chi^2 = 3.33$, df = 2 (P = 0.1892); $I^2 = 39.9\%$							
Test for overall effect: Z = 1.25 (P = 0.2122)							
Nausea							
Wu et al. 2024, (PACIFY COUGH)	6	43	3	43	39.3%	2.16	[0.50; 9.28]
Molyneaux et al. 2026 (CORAL)	52	125	2	40	39.2%	13.53	[3.13; 58.61]
Maher et al. 2023	16	42	0	42	21.5%	52.92	[3.05; 919.54]
Total (95% CI)	**74**	**210**	**5**	**125**	**100.0%**	**8.83**	**[1.62; 48.04]**
Heterogeneity: $Tau^2 = 1.3486$; $Chi^2 = 5.21$, df = 2 (P = 0.0740); $I^2 = 61.6\%$							
Test for overall effect: Z = 2.52 (P = 0.0118)							
Vomiting							
Maher et al. 2023	7	42	5	42	57.4%	1.48	[0.43; 5.10]
Wu et al. 2024, (PACIFY COUGH)	2	43	1	43	23.8%	2.05	[0.18; 23.48]
Molyneaux et al. 2026 (CORAL)	26	125	0	40	18.8%	21.57	[1.28; 362.50]
Total (95% CI)	**35**	**210**	**6**	**125**	**100.0%**	**2.64**	**[0.69; 10.10]**
Heterogeneity: $Tau^2 = 0.4152$; $Chi^2 = 2.91$, df = 2 (P = 0.2332); $I^2 = 31.3\%$							
Test for overall effect: Z = 1.42 (P = 0.1547)							
Constipation							
Maher et al. 2023	11	42	2	42	62.0%	7.10	[1.46; 34.38]
Molyneaux et al. 2026 (CORAL)	25	125	0	40	19.4%	20.55	[1.22; 345.67]
Wu et al. 2024, (PACIFY COUGH)	9	43	0	43	18.6%	23.96	[1.35; 426.22]
Total (95% CI)	**45**	**210**	**2**	**125**	**100.0%**	**10.94**	**[3.16; 37.89]**
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.77$, df = 2 (P = 0.6820); $I^2 = 0\%$							
Test for overall effect: Z = 3.77 (P = 0.0002)							
Somnolence							
Wu et al. 2024, (PACIFY COUGH)	4	43	2	43	41.5%	2.10	[0.36; 12.14]
Molyneaux et al. 2026 (CORAL)	14	125	1	40	30.0%	4.92	[0.63; 38.64]
Maher et al. 2023	9	42	1	42	28.5%	11.18	[1.35; 92.81]
Total (95% CI)	**27**	**210**	**4**	**125**	**100.0%**	**4.37**	**[1.41; 13.51]**
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 1.44$, df = 2 (P = 0.4870); $I^2 = 0\%$							
Test for overall effect: Z = 2.56 (P = 0.0105)							
Fatigue							
Molyneaux et al. 2026 (CORAL)	16	125	3	40	47.9%	1.81	[0.50; 6.57]
Maher et al. 2023	12	42	3	42	43.6%	5.20	[1.35; 20.09]
Wu et al. 2024, (PACIFY COUGH)	2	43	0	43	8.5%	5.24	[0.24; 112.45]
Total (95% CI)	**30**	**210**	**6**	**125**	**100.0%**	**3.14**	**[1.28; 7.69]**
Heterogeneity: $Tau^2 = 0.0040$; $Chi^2 = 1.34$, df = 2 (P = 0.5107); $I^2 = 0\%$							
Test for overall effect: Z = 2.50 (P = 0.0123)							
Dizziness							
Molyneaux et al. 2026 (CORAL)	23	125	2	40	72.7%	4.28	[0.96; 19.05]
Maher et al. 2023	10	42	0	42	27.3%	27.46	[1.55; 486.07]
Total (95% CI)	**33**	**167**	**2**	**82**	**100.0%**	**7.11**	**[1.40; 35.97]**
Heterogeneity: $Tau^2 = 0.3612$; $Chi^2 = 1.26$, df = 1 (P = 0.2608); $I^2 = 20.9\%$							
Test for overall effect: Z = 2.37 (P = 0.0177)							
Headache							
Maher et al. 2023	5	42	5	42	48.5%	1.00	[0.27; 3.75]
Molyneaux et al. 2026 (CORAL)	17	125	3	40	51.5%	1.94	[0.54; 7.00]
Total (95% CI)	**22**	**167**	**8**	**82**	**100.0%**	**1.41**	**[0.56; 3.53]**
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.5$, df = 1 (P = 0.4801); $I^2 = 0\%$							
Test for overall effect: Z = 0.73 (P = 0.4672)							
Dry mouth							
Maher et al. 2023	5	42	1	42	62.7%	5.54	[0.62; 49.63]
Molyneaux et al. 2026 (CORAL)	13	125	0	40	37.3%	9.72	[0.56; 167.27]
Total (95% CI)	**18**	**167**	**1**	**82**	**100.0%**	**6.83**	**[1.20; 38.79]**
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.09$, df = 1 (P = 0.7591); $I^2 = 0\%$							
Test for overall effect: Z = 2.17 (P = 0.0301)							

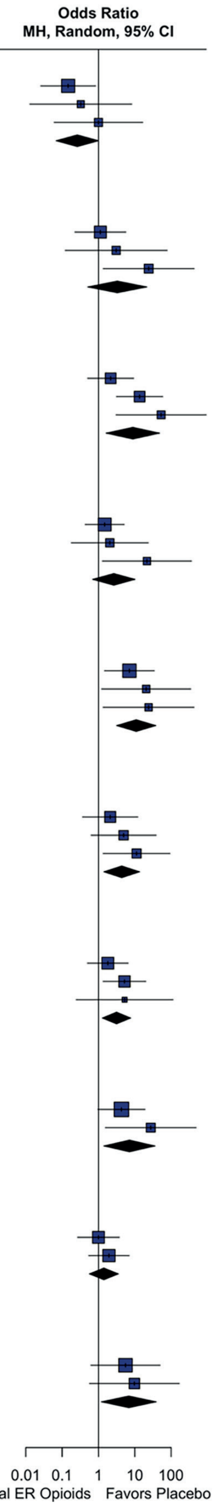

Figure 5. Forest plot for safety outcomes.

estimate was borderline and imprecise (OR 0.26; 95% CI 0.07 to 1.00; $p = 0.0501$; $I^2 = 0\%$). No statistically significant differences between ER oral opioids and placebo were observed for adverse events leading to discontinuation (OR 3.30; 95% CI 0.51 to 21.51; $p = 0.212$; $I^2 = 39.9\%$). Compared with placebo, ER oral opioids was associated with a higher incidence of nausea (OR 8.83; 95% CI 1.62 to 48.04; $p = 0.012$; $I^2 = 61.6\%$), constipation (OR 11.52; 95% CI 3.31 to 40.05; $p < 0.001$; $I^2 = 0\%$), somnolence (OR 4.37; 95% CI 1.41 to 13.51; $p = 0.011$; $I^2 = 0\%$), fatigue (OR 3.14; 95% CI 1.28 to 7.69; $p = 0.012$; $I^2 = 0\%$), dizziness (OR 7.11; 95% CI 1.40 to 35.97; $p = 0.018$; $I^2 = 20.9\%$), dry mouth (OR 6.83; 95% CI 1.20 to 38.79; $p = 0.030$; $I^2 = 0\%$). No significant differences between ER oral opioids and placebo were observed for vomiting (OR 2.64; 95% CI 0.69 to 10.10; $p = 0.16$; $I^2 = 31.3\%$) or headache (OR 1.41; 95% CI 0.56 to 3.53; $p = 0.47$; $I^2 = 0\%$).

Sensitivity analyses are presented in Supplementary Figures 3–8. All results were consistent with the primary analysis, except for the vomiting endpoint (Supplementary Figure 8D). After a correlation of 0.75, ER oral opioids increase the chance of vomiting (OR: 1.74; 95%CI: 1.00 to 3.03).

Risk of bias assessment

Two RCTs (Maher 2023, et al and PACIFY COUGH trial) were judged to have some concerns regarding risk of bias, while one RCT (CORAL trial) was assessed as a low overall risk of bias on the RoB-2 risk assessment tool. Individual study appraisals using the RoB-2 tool are presented in Supplementary Figure 1.

GRADE assessment

The certainty of evidence was rated as moderate for objective 24-hour cough frequency, E-RS IPF cough subscale score, LCQ score, dizziness, headache, and dry mouth; low for CS-NRS score, serious adverse events, nausea, vomiting, constipation, somnolence, and fatigue; and very low for adverse events leading to discontinuation (Supplementary Figure 2).

For 24-hour cough frequency, CS-NRS, and E-RS scale, the certainty of evidence for efficacy outcomes was downgraded because of substantial heterogeneity across studies. For CS-NRS and LCQ score, the certainty of evidence for efficacy outcomes was downgraded for imprecision because the 95% CI crossed the MCID. For serious adverse events, the certainty of evidence was downgraded due to risk of bias in the included studies and imprecision, as few events were observed and the confidence interval was wide. The certainty of evidence for adverse events leading to discontinuation was downgraded due to serious imprecision, as the confidence interval was very wide and included both potential benefit and harm, with few events.

Discussion

In this systematic review and meta-analysis of three RCTs with 250 patients with IPF, ER oral opioid therapies significantly improved objective 24-hour cough frequency, CS-NRS score, and E-RS IPF cough subscale score compared with placebo. The efficacy outcomes were derived solely from trials evaluating extended-release nalbuphine. Rates of serious adverse events and adverse events leading to discontinuation were comparable between ER oral opioids and placebo, although non-serious adverse events such as nausea, constipation, fatigue, dizziness, and dry mouth were more frequent with ER oral opioid therapy.

Cough imposes a substantial burden in patients with IPF (4). A previous study showed that worsening cough-related quality-of-life scores were associated with increased risks of respiratory-related hospitalization, death, and lung transplantation independent of disease severity (26). Studies of refractory chronic cough may offer insights into the treatment of cough in IPF. However, differing results from trials of the peripherally acting P2X3 receptor antagonist gefapixant suggest that the biological mechanisms contributing to cough may not be the same in these conditions (27,28). To date, few clinically useful therapies have demonstrated clear efficacy specifically for cough in IPF, and no established treatment is currently available (4).

In the present meta-analysis, ER oral opioids likely improved objective 24-hour cough frequency. This outcome is clinically important because objective 24-hour cough frequency provides a quantitative and reproducible measure for assessing treatment effects and is incorporated into cough severity and quality-of-life instruments (29). Evidence that other pharmacologic therapies improve objective cough frequency in IPF remains limited. In an observational study, pirfenidone was associated with a reduction in objective cough frequency; however, no comparison with a control group was available (30). In addition, the phase 2b trial of inhaled cromolyn sodium did not demonstrate a statistically significant difference compared with placebo (11). Consequently, the clinical improvement observed with ER nalbuphine relative to placebo in this study suggests it may represent a viable therapeutic intervention for cough in the IPF population. The fact that the observed reduction exceeded the proposed meaningful change threshold further supports the potential clinical relevance of this finding.

However, moderate heterogeneity was observed across studies for primary outcome. This heterogeneity can be explained by differences in dosing regimens. Maher et al. (2023) used a titration regimen ranging from 27 to 162 mg twice daily, whereas the CORAL trial evaluated three fixed doses (27, 54, and 108 mg twice daily), potentially influencing treatment effects across studies (15,16). With the titration regimen, the geometric mean difference showed a greater reduction in cough frequency compared with the combined estimates from the fixed-dose groups (15,16). In addition, within the CORAL trial, higher fixed doses were associated with greater reductions in cough frequency (16).

In chronic cough, the correlation between objective cough frequency and patient-centered outcomes has been reported to be only moderate (31). Incorporating patient-reported measures when evaluating treatment efficacy is necessary to capture meaningful improvements in patients' symptoms (29). In the present study, ER oral opioids may modestly improve patient-reported cough severity assessed by the CS-NRS, likely improve subjective cough frequency assessed by the E-RS IPF cough subscale and cough-related quality of life assessed by the LCQ. Moreover, the E-RS IPF cough subscale score has been reported to be associated with health-related quality of life measures such as the St. George's Respiratory Questionnaire (32). These findings suggest that ER oral opioids may have favorable effects on cough-related quality-of-life outcomes, in addition to reducing objective cough frequency. Similar to primary outcome, the heterogeneity may also be attributable to differences in dosing regimens.

For safety outcomes, previous studies of ER morphine in patients with chronic obstructive pulmonary disease have reported serious adverse events in up to 33% of patients (33). This suggested that opioid use in progressive respiratory diseases requires careful consideration and close monitoring. There was no clear evidence of an increased risk of serious adverse events compared with placebo. However, given the borderline p-value and the upper limit of the 95% CI reaching 1.00, this finding should be interpreted cautiously, as it may reflect limited statistical power due to the small number of events. Regarding minor adverse effects, a trial of slow-release morphine sulfate for chronic cough reported constipation and somnolence as the most common adverse events (34). Similarly, the present analysis showed that ER oral opioid therapy likely increased dizziness and dry mouth, and may increase vomiting. Although nalbuphine functions as a κ-opioid receptor agonist and μ-opioid receptor antagonist, which may be associated with a lower risk of respiratory depression (35), respiratory depression was not specifically reported as an outcome in the included RCTs and therefore could not be pooled in the meta-analysis. Collectively, these findings suggest that, with appropriate monitoring for adverse effects, ER oral opioids may be cautiously considered as a therapeutic option for cough in patients with IPF.

This meta-analysis pooled clinically relevant efficacy and safety outcomes of only RCTs, providing more robust evidence for the use of ER oral opioid therapy in this population. In previous RCTs, safety outcomes were reported only as event counts; by synthesizing these data, our analysis was able to quantitatively evaluate differences in adverse events between treatment groups (5,15,16).

Limitations of this study should be acknowledged. First, the number of included trials was limited, which may reduce the precision of pooled estimates. Second, the median follow-up duration across the included trials was 56 days; therefore, the long-term safety of ER oral opioids, including tolerance and physical dependence, warrants further investigation. Third, the certainty of evidence for several safety outcomes was rated as low or very low according to the GRADE framework. Forth, the patient-reported outcomes should be interpreted cautiously. Although ER oral opioids significantly improved the CS-NRS and E-RS IPF cough subscale scores, the CS-NRS did not clearly reach the reference threshold for clinical meaningfulness. Finally, the crossover design and the absence of individual patient data precluded a more granular analysis, potentially limiting the generalizability of our findings.

In conclusion, ER oral opioids likely reduce objective and subjective cough frequency and likely improve cough-related quality of life in patients with IPF, without clear evidence of an increased risk of serious adverse events or treatment discontinuation. In patients with IPF who experience severe cough symptoms, ER oral opioids may be considered with careful monitoring for adverse effects.

Ethics Approval: The authors have nothing to report.

Conflict of Interest: The authors declare no conflict of interest.

Authors' Contribution: S.N. contributed to study conception and design, literature search, study selection, data extraction, data curation, statistical analysis, risk of bias assessment, certainty assessment, interpretation of the findings, figure preparation, and drafted the manuscript. A.V. contributed to literature search, study selection, risk of bias assessment, certainty assessment, interpretation of the findings, and critically revised the manuscript. M.M. contributed to data extraction, data curation, interpretation of the findings, and critically revised the manuscript. B.M. contributed to study conception and design, interpretation of the findings, supervised the study, and critically revised the manuscript. All authors read and approved the final manuscript and agree to be accountable for all aspects of the work.

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Appendix – Supplementary files

Supplementary data can be found online.